



DAY 15: SPREAD (...) AND REST (...) OPERATORS IN JAVASCRIPT



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JavaScript's spread (...) and rest (...) operators are two powerful features introduced in ES6 that make handling arrays, objects, and function parameters much easier.

Spread Operator (...)

The spread operator (...) expands elements of an array, object, or iterable into individual elements. It's mainly used for:

- ✓ Copying arrays/objects
- ✓ Merging arrays/objects
- ✓ Passing elements as function arguments

✓ Example 1: Copying an Array

```
const numbers = [1, 2, 3];  
const newNumbers = [...numbers];  
  
console.log(newNumbers);  
// Output: [1, 2, 3]  
console.log(numbers === newNumbers);  
// Output: false (different memory  
reference)
```

Why use spread? Direct assignment (=) creates a reference,
but ... creates a new copy.

✓ Example 2: Merging Arrays

```
const arr1 = [1, 2, 3];  
const arr2 = [4, 5, 6];  
const mergedArr = [...arr1, ...arr2];  
  
console.log(mergedArr);  
// Output: [1, 2, 3, 4, 5, 6]
```

- ◆ Much cleaner than concat()!

✓ Example 3: Spreading in Function Arguments

```
function sum(a, b, c) {  
  return a + b + c;  
}
```

```
const numbers = [1, 2, 3];  
console.log(sum(...numbers)); // Output: 6
```

- ◆ Replaces apply() method and makes code simpler!

✓ Example 4: Copying & Merging Objects

```
const person = { name: "Alice", age: 25 };  
const newPerson = { ...person, city: "New  
York" };  
console.log(newPerson);  
// Output: { name: "Alice", age: 25, city:  
"New York" }
```

- ◆ Spread is useful for shallow copying objects without modifying the original.

Rest Operator (...)

The rest operator (...) collects multiple arguments or properties into a single array or object.

It is mainly used in:

- ✓ Function parameters to accept multiple arguments
 - ✓ Destructuring arrays and objects

✓ Example 1: Rest in Function Parameters

```
function sum(...numbers) {  
  return numbers.reduce((acc, num) => acc +  
    num, 0);  
}  
console.log(sum(1, 2, 3, 4, 5));  
// Output: 15
```

- ◆ Instead of passing individual parameters, we capture all into an array.

✓ Example 2: Rest with Destructuring (Arrays)

```
const [first, second, ...remaining] =  
[10, 20, 30, 40, 50];  
console.log(first); // Output: 10  
console.log(second); // Output: 20  
console.log(remaining); // Output:  
[30, 40, 50]
```

- ◆ The rest operator groups remaining values into remaining.

✓ Example 3: Rest with Object Destructuring

```
const person = { name: "Alice", age: 25, city: "New York", country: "USA" };
```

```
const { name, age, ...location } = person;
```

```
console.log(name); // Output: Alice  
console.log(age); // Output: 25  
console.log(location); // Output: {  
  city: "New York", country: "USA" }
```

- ◆ location holds all remaining properties after name and age.

Differences Between Spread and Rest Operators

Feature	Spread Operator (...)	Rest Operator (...)
Usage	Expands elements into individual items	Gathers multiple elements into a single array
Used In	Arrays, Objects, Function Calls	Function Parameters, Destructuring
Example	<code>const newArr = [...arr]</code>	<code>function sum(...args) {}</code>

